

DAYLIFF 1000 EYES

Enterprise Process Observability, Workflow Intelligence and Operational Analytics Platform

1. EXECUTIVE SUMMARY

Dayliff 1000 Eyes is an enterprise-grade operational intelligence platform designed to provide complete visibility into the lifecycle of customer requests across multiple departments and systems.

The platform serves as a centralized observability layer that sits above existing business applications such as CRM systems, ERP platforms, engineering software, quotation systems, logistics systems, and customer support tools.

Instead of replacing existing infrastructure, Dayliff 1000 Eyes integrates with those systems through APIs, webhooks, event streams, and manual inputs to create a unified real-time representation of the customer journey.

The system continuously monitors operational workflows, reconstructs customer journeys, measures SLA compliance, identifies bottlenecks, predicts delays using artificial intelligence, and provides decision-makers with actionable insights.

The ultimate objective is to eliminate operational blind spots, improve accountability, reduce delays, enhance customer experience, and provide predictive operational intelligence.

2. BUSINESS PROBLEM

Current operational processes are fragmented across multiple independent systems.

A customer request may pass through:

Customer → Sales → Engineering → Quotation → Approval → Logistics → Delivery

However, each department uses different tools and databases.

This creates:

- Process visibility gaps
- Delayed handoffs

- Poor accountability
- SLA breaches
- Customer frustration
- Manual tracking overhead
- Difficulty measuring performance
- Inability to identify bottlenecks

Management lacks a single source of truth that shows:

- Current request status
 - Ownership
 - Delay reasons
 - Department performance
 - SLA compliance
 - Customer journey progress
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3. PROJECT OBJECTIVES

The system shall:

1. Provide end-to-end visibility of customer requests.
 2. Track requests in real time.
 3. Reconstruct customer journeys.
 4. Monitor SLA compliance.
 5. Detect bottlenecks automatically.
 6. Generate alerts for delayed processes.
 7. Produce operational analytics.
 8. Provide role-based dashboards.
 9. Support AI-powered predictions.
 10. Act as a centralized observability platform.
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4. HIGH LEVEL ARCHITECTURE

The platform consists of six major layers:

1. Core Domain Layer
 2. Event Processing Layer
 3. Workflow Intelligence Layer
 4. API Layer
 5. Real-Time Communication Layer
 6. AI Intelligence Layer
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5. CORE DOMAIN LAYER

The Core Domain Layer contains the primary business entities.

Customer

Represents a client.

Attributes:

- Customer ID
 - Name
 - Email
 - Phone Number
 - Region
 - Organization
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Service Request

Represents a customer request.

Attributes:

- Request ID
- Request Number
- Customer ID
- Priority
- Current Stage
- Current Status
- Assigned Department
- Created At
- Updated At

Examples:

- Borehole Design Request
 - Solar Installation Request
 - Water Pump Consultation
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Journey Stage

Represents each step within the request lifecycle.

Examples:

- Inquiry
 - Sales Review
 - Engineering Design
 - Quotation
 - Approval
 - Dispatch
 - Delivery
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Event

An immutable record of an activity.

Examples:

- REQUEST_CREATED
- SALES_ASSIGNED
- DESIGN_STARTED
- DESIGN_COMPLETED
- QUOTATION_GENERATED
- DELIVERY_COMPLETED

Each event contains:

- Event ID
 - Request ID
 - Event Type
 - Timestamp
 - Department
 - User
 - Source System
 - Metadata
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SLA Rule

Defines acceptable processing times.

Example:

Engineering Design: Maximum Duration = 24 Hours

Alert

Represents warnings and SLA breaches.

Severity Levels:

- Info
 - Warning
 - Critical
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Notification

Represents messages delivered through:

- Dashboard
 - Email
 - SMS
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6. EVENT-DRIVEN ARCHITECTURE

The platform follows an event-driven design.

Every meaningful business activity becomes an event.

Example:

Customer submits request.

System generates:

REQUEST_CREATED

The event enters the Event Processing Pipeline.

Pipeline:

Receive Event → Validate Event → Store Event → Update Request State → Update Timeline → Evaluate SLA
→ Generate Alerts → Broadcast Updates → Update Analytics

Benefits:

- Real-time processing

- Traceability
 - Auditability
 - Scalability
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7. EVENT PROCESSING LAYER

Responsibilities:

- Receive events
- Validate payloads
- Normalize external data
- Store events
- Trigger workflows

Supported Sources:

CRM ERP Engineering Software Quotation Systems Logistics Platforms Manual Input

Integration Methods:

REST APIs Webhooks Scheduled Synchronization

8. CUSTOMER JOURNEY RECONSTRUCTION

One of the platform's most important features.

The system reconstructs customer journeys using events.

Example Timeline:

09:00 Request Created 09:15 Sales Assigned 10:00 Engineering Started 13:00 Design Completed 14:00 Quotation Generated 15:00 Approval Received 16:00 Dispatch Started

This creates complete visibility.

Benefits:

- Traceability
 - Accountability
 - Historical auditing
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9. SLA MONITORING ENGINE

The SLA Engine continuously monitors workflows.

Responsibilities:

- Track stage durations
- Compare against SLA thresholds
- Generate alerts
- Trigger escalations

Example:

Engineering Design SLA: 24 Hours

If duration exceeds threshold:

Generate Alert Notify Stakeholders Update Dashboard

10. ALERT MANAGEMENT SYSTEM

Alert Types:

Operational Alert SLA Breach Alert Critical Delay Alert Escalation Alert

Alert Channels:

Dashboard Notification Email SMS

Alert Lifecycle:

Created → Acknowledged → Resolved → Archived

11. ANALYTICS ENGINE

The Analytics Engine transforms operational data into insights.

Metrics:

Average Completion Time Average Department Processing Time SLA Compliance Rate Request Throughput
Delay Frequency Completion Rate

Reports:

Daily Weekly Monthly Quarterly

12. ARTIFICIAL INTELLIGENCE LAYER

The AI Layer transforms the platform from monitoring to prediction.

Level 1 AI

Predictive Analytics

Capabilities:

Delay Prediction SLA Breach Prediction Risk Scoring Completion Time Prediction

Example:

Request Probability of Delay: 82%

Risk Level: High

Level 2 AI

Operational Copilot

Natural Language Queries

Examples:

"Show all delayed requests."

"Which department causes most delays?"

"What is today's SLA compliance?"

AI translates questions into analytics queries.

13. REAL-TIME COMMUNICATION

Technology:

WebSockets Redis Pub/Sub

Purpose:

Instant updates without page refresh.

Features:

Live Dashboard Updates Live Alerts Live Analytics Refresh Live Status Changes

14. ROLE-BASED ACCESS CONTROL

Roles:

Administrator Regional Manager Sales Engineer Backend Designer Logistics Officer

Each role receives:

Different permissions Different dashboards Different analytics visibility

15. API ARCHITECTURE

Authentication APIs

POST /auth/login POST /auth/logout POST /auth/refresh

Request APIs

GET /requests GET /requests/{id} POST /requests PATCH /requests/{id}

Event APIs

POST /events GET /events GET /events/{id}

Timeline APIs

GET /timeline/{request_id}

Dashboard APIs

GET /dashboard/overview GET /dashboard/kpis GET /dashboard/bottlenecks GET /dashboard/performance

Analytics APIs

GET /analytics/trends GET /analytics/sla GET /analytics/efficiency

Alert APIs

GET /alerts PATCH /alerts/{id}

AI APIs

POST /ai/risk-score POST /ai/predict-delay POST /ai/copilot

16. DATABASE DESIGN

Core Tables:

users roles departments customers service_requests journey_stages events sla_rules alerts notifications analytics_snapshots

Database Engine:

PostgreSQL

Reasons:

Reliability Performance Scalability JSON Support

17. TECHNOLOGY STACK

Backend: FastAPI

Database: PostgreSQL

Caching: Redis

Background Jobs: Celery

ORM: SQLAlchemy

Authentication: JWT

Documentation: OpenAPI / Swagger

Deployment: Docker

Container Orchestration: Docker Compose

18. UI/UX DESIGN PRINCIPLES

Color Palette:

Black White Cyan

Design Philosophy:

Enterprise Minimalist Modern Professional

Human Computer Interaction Principles

Visibility of System Status

Users must always know:

Current Stage Current Status Remaining SLA

Consistency

Uniform layouts Uniform navigation Uniform interactions

Recognition Over Recall

Users should not memorize information.

Everything must be visible.

Progressive Disclosure

High-level overview first.

Detailed data on demand.

Error Prevention

Validation Warnings Confirmations

Accessibility

Readable typography High contrast Keyboard navigation Responsive layouts

19. DASHBOARD MODULES

Executive Dashboard

KPI Overview Active Requests Critical Alerts SLA Compliance

Operations Dashboard

Workflow Monitoring Timeline Tracking Department Performance

Analytics Dashboard

Trends Reports Predictions

AI Dashboard

Predictions Risk Analysis Recommendations

20. DEVELOPMENT ROADMAP

Phase 1

Project Setup

Repository Docker Database Authentication

Phase 2

Core Domain Models

Users Requests Events Stages Alerts

Phase 3

Event Processing

Webhooks Event Pipeline Timeline Builder

Phase 4

Real-Time Layer

WebSockets Redis

Phase 5

Analytics

KPIs Reports Metrics

Phase 6

AI Layer

Predictions Copilot

Phase 7

UI/UX

Dashboards Journey Visualization

Phase 8

Testing

Unit Tests Integration Tests Load Testing

Phase 9

Deployment

Production Environment Monitoring Logging

21. FUTURE EXPANSION

Mobile Application ERP Integrations IoT Device Monitoring Voice Assistants Advanced AI Agents Predictive Automation Enterprise Multi-Region Support

22. SUCCESS CRITERIA

The project is successful when:

Every customer request is traceable.

Every department is accountable.

Every SLA is measurable.

Every delay is visible.

Management has complete operational visibility.

AI predicts problems before they occur.

The platform becomes the operational intelligence layer for Dayliff.